

EIGRP Routing — AS 100

Setting	Router (RCA-RTR01)	Switch (RCA-SW01)
Networks	192.168.1.0/30 (transit only)	Transit /30 + all 6 VLAN subnets
Default Route	redistribute static metric 10000 100 255 1 1500 → pushes 0.0.0.0/0 to switch	Learns default dynamically from router via EIGRP
Static Routes	Only 0.0.0.0/0 → 10.0.0.1 remains	All removed — learned via EIGRP
Passive Interfaces	passive-interface default / active only on Gi0/1/0	passive-interface default / active only on Gi1/0/48
Authentication	MD5 key chain RCAEIGRPKEY on Gi0/1/0	Matching MD5 key chain on Gi1/0/48
Logging	eigrp log-neighbor-changes + log-neighbor-warnings	Same
SNMP Traps	snmp-server enable traps eigrp	Same

Syslog — End-to-End Verification

The syslog pipeline is fully wired:

Cisco Devices —UDP 1514—→ Promtail (syslog scrape) —→ Loki —→ Grafana

Component	Config Line	Status
Router syslog out	logging host 192.168.99.10 transport udp port 1514 / source Gi0/1/0	✓
Switch syslog out	logging host 192.168.99.10 transport udp port 1514 / source Vlan99	✓
Origin ID	logging origin-id hostname on both — Promtail uses this for host label	✓
Trap level	logging trap informational (level 6) — captures EIGRP, link, login, config events	✓
Promtail listener	0.0.0.0:1514/udp with syslog protocol, relabels hostname/severity/facility	✓
Cisco facility regex	Pipeline stage extracts %FACILITY-SEV-MNEMONIC as cisco_facility label	✓
Loki ingestion	Promtail pushes to http://loki:3100/loki/api/v1/push	✓

Updated Grafana Dashboard — 20 Panels / 5 Rows

Row	Panels
1	Device Health Router Uptime; Switch Uptime; Router CPU; Switch CPU; Router Mem %; Switch Mem %

|  EIGRP AS 100 | Router Neighbor Count; Switch Neighbor Count; Peer Retransmissions; Queued Pkts; Hold Time; Neighbor Flaps (24h from Loki); Router EIGRP Packet Rates chart; Switch EIGRP Packet Rates chart |

|  Interface Traffic | WAN Traffic (Gi0/0/0); Transit Link Traffic (labeled "EIGRP Adjacency") |

|  Errors & Port Status | Interface Errors bar chart; Switch Port Status table (UP/DOWN color-coded) |

|  Syslog | Router Syslog Health (5min count — red if 0); Switch Syslog Health; Ingestion Rate by Host; Volume by Severity; EIGRP Event Stream (filters EIGRP\|DUAL-\|NBRCHANGE); Link & Security Event Stream; Full Live Syslog Stream |

SNMP Exporter Update

Added CISCO-EIGRP-MIB (1.3.6.1.4.1.9.9.449) — the exporter now polls:

- Traffic stats: neighbor count, hellos/updates/queries/replies/ACKs sent & received
- Peer table: peer address, interface, hold time, uptime, retransmissions, queue depth

> Rename snmp-exporter-config.txt → snmp.yml when deploying.